

Technical specification for ARER-300

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RFQ-003 §1.1 — The platform shall test at least 48 specimens simultaneously in one controlled run. Every position must support synchronised temperature, mechanical load and electrical bias. Sequential batches are not equivalent to one run.

RFQ-003 §1.2 — Required programmable ranges are -40°C to 125°C , 0 to 500 N load and 0 to 24 V electrical bias. The platform records raw readings, event timestamps and fault events. Addendum ADD-01 clarified the lower temperature endpoint wording without changing the -40°C requirement.

RFQ-003 §2.1 — CSV or JSON export of raw measurements and event logs must operate without an ongoing vendor-hosted service. The supplier installs and commissions the platform, executes SAT and trains eight staff on site. Acceptance is evidenced by an SAT record.

RFQ-003 §3 Control and measurement architecture

RFQ-003 §3.1 — Each simultaneous specimen position should carry a unique channel identifier through the run record. The platform should time-align temperature, applied load, electrical bias and observed response so the research team can examine a failure event against the operating profile. The bidder should describe how setpoints, actual readings, warnings and interrupted-run events are captured. The SAT should demonstrate representative operation rather than rely solely on a brochure maximum.

RFQ-003 §3.2 — Bidders may offer more positions or additional analytics, but the required comparison is based on the delivered configuration at tendered price. A chamber that processes batches consecutively should report its simultaneous capacity plainly. Any expansion hardware needed to meet the minimum belongs in the evaluated configuration and price.

RFQ-003 §4 Data handling and acceptance evidence

RFQ-003 §4.1 — A usable export should preserve channel identifiers, timestamps, units, operating settings, raw readings and event codes in a documented CSV or JSON structure. The Institute should be able to retain and read the files locally after a hosted account ends. A proprietary dashboard may be offered as an additional convenience, provided it is not the sole means of obtaining the mandatory raw data and logs.

RFQ-003 §4.2 — At SAT, the supplier should show a configured run, export a sample file, reconcile selected values with the controller display and demonstrate an event log entry. The operator group should receive a short explanation of routine startup, safe shutdown, export and first-line fault reporting. Acceptance evidence should identify any deviations and the decision maker who closes them.

RFQ-003 §5 Documentation package

RFQ-003 §5.1 — The supplier should provide an as-installed configuration list, operating instructions, a description of exported fields and a procedure for retrieving event logs. The laboratory must be able to interpret units and timestamps after the installation team leaves.

Training should use the delivered configuration and include a hands-on export exercise so the operators know which file is the raw record and which is a convenience report.

RFQ-003 §5.2 — Any deviation observed at SAT should be entered in the signed record with an owner and resolution route. A later correction may be accepted through the contract process, but the tender evaluation must assess the commitment in the submitted plan and technical schedule. Vendor demonstrations and brochures may supplement the record but do not replace the specific tendered configuration.

Fictional training case; all people, organisations, products, rules and events are invented. This is not A*STAR policy or Singapore procurement law.